

Over The Air Computing

Master Thesis

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April 2026

Abstract – Over-the-Air (OTA) is an emerging technique that leverages the superposition property of wireless channels to perform computations directly during the transmission process. This approach has the potential to significantly reduce latency and energy consumption in distributed systems, making it particularly suitable for applications in Internet of Things (Internet of Things (IoT)) networks, edge computing, and real-time data processing. This thesis explores the theoretical foundations of OTA, including channel modeling, signal processing techniques, and error analysis. Practical implementations and case studies demonstrating the effectiveness of OTA in various scenarios are also presented.

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List of Acronyms and Abbreviations

AWGN	Additive White Gaussian Noise 3–6, 9, 10
CSI	Channel State Information 4, 16
DSP	Digital Signal Processing 10
ED	End Device 4, 13, 16
ES	Edge Server 13
FC	Fusion Center 4, 13, 14
FIR	Finite Impulse Response 10–12
IoT	Internet of Things 1
ISI	Inter-Symbol Interference 3, 4, 14, 16–18
MAC	Multiple Access Channel 4, 12
MIMO	Multiple-Input Multiple-Output 19
MSE	Mean Squared Error 5, 6, 9, 11, 14–16
OTA	Over-the-Air 1, 3, 4, 8, 13–15, 17
PAM	Pulse Amplitude Modulation 4, 9, 13
RC	Raised Cosine 5
RRC	Root Raised Cosine 5, 6, 8, 10, 11
SDR	Software Defined Radio 3, 10
sps	Symbols per Sample 5–8, 10, 11
TDMA	Time-Division Multiple Access 19
α	Roll-off Factor 5, 6

Introduction

Over the Air Computing offers a paradigm shift in how distributed computations are performed in wireless networks. By exploiting the inherent properties of the wireless medium, OTA enables simultaneous transmissions from multiple devices to be combined in a way that directly computes a desired function of the transmitted data. This technique has garnered significant attention due to its potential to enhance efficiency and reduce latency in applications such as federated learning, distributed sensing, and real-time analytics. The motivation behind this thesis is to explore the practical implementation of OTA using Software Defined Radios Software Defined Radio (SDR) and MATLAB, addressing the challenges associated with synchronization, channel impairments, and waveform design. The research objectives include designing a prototype OTA network, validating its performance, and analyzing the impact of various impairments on computation accuracy.

0.a Background and Motivation

The increasing need for efficiency in computation-oriented applications where the interest lies in a function of local information, rather than the information itself [6]. The conventional approach of separating communication and computation often proves suboptimal [3], [4], [6]. The potential of OTA Computation is to achieve significantly higher computation rates by harnessing interference through signal superposition in a wireless Multiple Access Channel (MAC) [6].

0.b Problem Statement and Research Objectives

Focusing on designing and validating a practical OTA network layout suitable for implementation in a controllable environment[6]. The specific role of Software Defined Radios (SDRs) and MATLAB/simulation tools for prototype validation and addressing practical limitations such as synchronization and channel impairments [2], [6], [7].

0.c Thesis Contributions and Organization

The aim of this thesis is to contribute to the research objectives...as a student member of the university of western macedonia, department of electrical and computer engineering, under the supervision of professor

1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading, and time synchronization errors, it is essential to establish a baseline model of OTA computation under ideal conditions. The foundational premise of OTA computing is its ability to reduce the communication latency of data aggregation from the linear complexity shown in Eq. (1) to the constant complexity made possible by allowing K devices to transmit simultaneously [6].

$$\mathcal{O}(K) \longrightarrow \mathcal{O}(1) \quad (1)$$

In this idealized scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted solely by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, algorithmic MATLAB modeling, block-level Simulink architecture, and GNU Radio flowgraphs that demonstrate this ideal, unimpaired baseline. Establishing this benchmark is critical for quantifying the exact degradation caused by physical channel constraints in subsequent chapters.

1.a Mathematical Formulation of the Ideal Channel

Consider a distributed wireless sensor network consisting of K End Device (ED)s transmitting local analog values to a central Fusion Center (FC). Let $x_k(t)$ denote the continuous-time, real-valued signal transmitted by the k -th device. For analog OTA computation, uncoded Pulse Amplitude Modulation (PAM) is widely considered the optimal transmission scheme, as it linearly maps the sensor readings directly to the amplitude of the transmitted waveform [4], [9].

In an ideal Multiple Access Channel (MAC), we assume perfect Channel State Information (CSI) at the transmitters, which allows for flawless pre-equalization of phase shifts and path loss attenuation. Consequently, the electromagnetic waves superimpose perfectly at the receiving antenna of the FC. The received baseband signal $y(t)$ is mathematically expressed as the linear sum of the transmitted signals plus noise:

$$y(t) = \sum_{k=1}^K x_k(t) + n(t) \quad (2)$$

where the AWGN model is specified by Eq. (3). Because there is no multipath delay spread in this baseline model, the channel impulse response is effectively the Dirac delta function in Eq. (4). This implies that the current symbol is entirely unaffected by past or future symbols, resulting in zero ISI. To extract the arithmetic average of the K sensor readings, the FC simply applies a post-processing scaling factor of $1/K$ to the received aggregated signal $y(t)$.

$$n(t) \sim \mathcal{N}(0, \sigma^2) \quad (3)$$

$$h(t) = \delta(t) \quad (4)$$

1.b MATLAB Simulation and Code Analysis

To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

```

1  % Ideal OTA Computation Simulation
2  K = 3; % Number of sensors
3  N = 1000; % Number of samples
4  SNR_dB = 20; % Signal-to-Noise Ratio
5
6  % Generate random temperature readings around 27 degrees
7  X = 27 + randn(K, N);
8
9  % The Air acts as an ideal adder (MAC Superposition)
10 Ideal_Sum = sum(X, 1);
11
12 % Add AWGN
13 Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');
14
15 % Calculate the Fusion Center Average
16 Estimated_Average = Received_Signal / K;

```

```

17
18 % Display Mean Squared Error (MSE)
19 MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE
    at %d dB SNR: %f\n', SNR_dB, MSE);

```

While Listing 1 demonstrates the fundamental principle in its unadulterated form, practical wireless systems necessitate pulse shaping to satisfy spectral constraints and facilitate signal recovery. Consequently, a more physically realistic model incorporates matched Raised Cosine (RC) filtering at both the transmitter and receiver. This architectural refinement introduces additional degrees of freedom—namely, the Symbols per Sample (sps) parameter and the roll-off factor (α)—which materially influence system performance. The enhanced implementation is presented in Listing 2.

Listing 2: MATLAB script for OTA computation aggregation using the Root Raised Cosine (RRC) filter.

```

1 txUpSamped = upsample(X', sps_val); % phase may need to be set
2
3 % Raised Cosine Filter
4 rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");
5
6 rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);
7
8 for k = 1:K
9     rx_agg_signal = rx_agg_signal + conv(txUpSamped(:,k), rcFilter(:));
10 end
11
12 Received_Signal = awgn(rx_agg_signal, current_snr, 'measured');
13
14 matchedSignal = conv(Received_Signal, flipud(rcFilter(:)), 'same');
15
16 rxMatched = downsample(matchedSignal, sps_val);
17
18 start_idx = floor(span/2); stop_idx = floor(span/2) + mod(span,2) + 1;
19
20 % Truncate edges to avoid filter transient errors
21 Estimated_Average = rxMatched(start_idx:end-stop_idx, :) / K;
22
23 % Calculate and store MSE in the matrix
24 mse_results(snr_idx, a_idx) = immse(Estimated_Average', mean(X, 1));

```

As demonstrated by the script, when synchronization is perfect, the MSE is strictly a function of the AWGN variance. When however trying to implement a scenario that is closer to a real world usecase, it appears to be necessary to interject a filter pair like the raised cosine filter. In order for this to be accomplished, a matched root raised cosine filter shall be inserted on each end of channel and by doing that, more degrees of freedom shall be taken into account, like sps and Roll-off Factor (α). In diagram 1b the curve can be seen where in diagram 1a the ideal Mean Squared Error (MSE) can be shown.

Computational execution of the aforementioned algorithms reveals a critical distinction between the idealized and practical regimes. When temporal synchronization is mathematically perfect and no filtering is applied, the estimation accuracy is entirely governed by the noise floor; the MSE exhibits a direct proportionality to the AWGN variance and inverse quadratic scaling with the number of transmitting sensors. Mathematically, this relationship is expressed in Eq. (5), which demonstrates

the fundamental advantage of cooperative multi-device aggregation [1], [6]. Conversely, when pulse shaping filters are incorporated into the transmission chain, the interplay between symbol timing parameters (sps) and spectral efficiency (α) introduces measurable deviation from the ideal theoretical bound.

$$MSE \propto \frac{\sigma^2}{K^2} \quad (5)$$

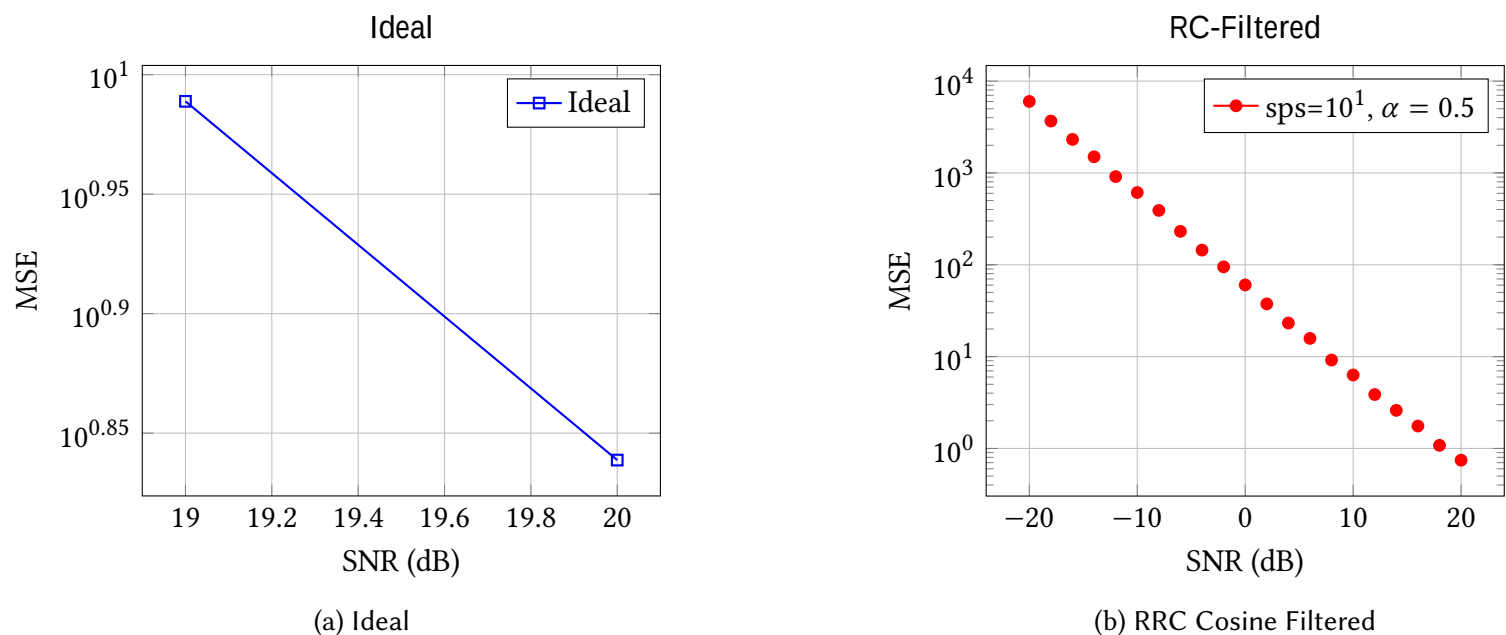


Figure 1: Comparison of MSE performance between the ideal baseline and the Raised Cosine filtered scenario.

Figure 1a presents the theoretical performance floor, wherein the MSE is a function solely of AWGN variance. In contrast, Figure 1b illustrates the performance envelope when matched Raised Cosine filters are deployed at both transmitter and receiver, with sps and α held constant. Despite the introduction of filtering, the channel remains free of multipath distortion and timing misalignment, ensuring that noise contributions remain the dominant performance-limiting factor.

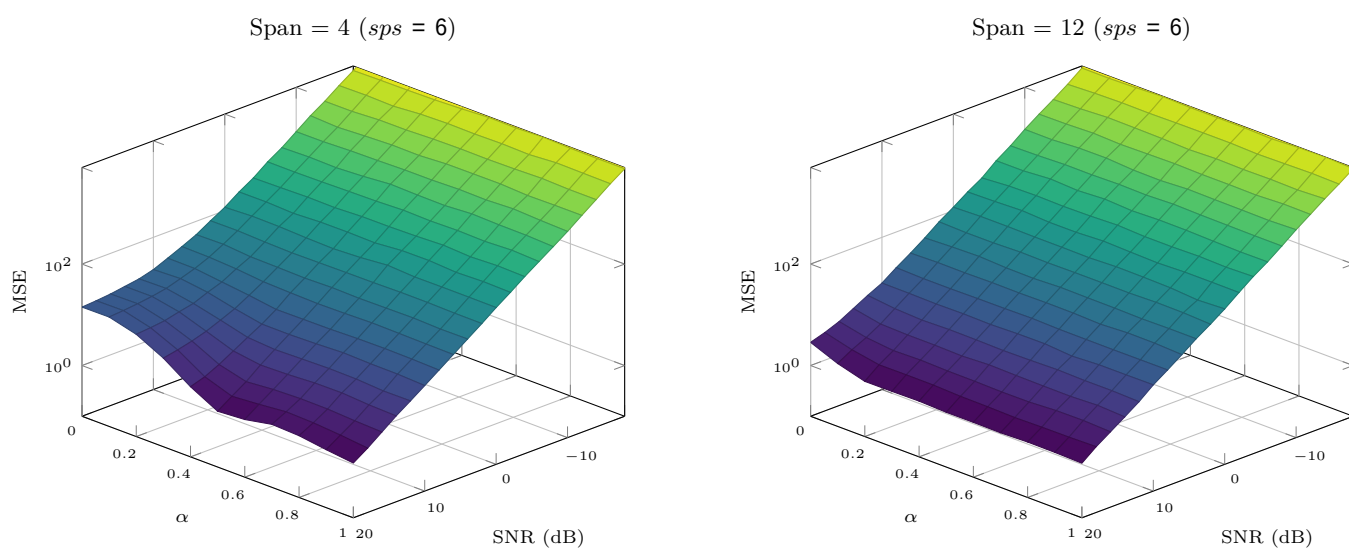


Figure 2: Synchronous 3D animation comparing the Over-The-Air MSE surface across sps scaling for filter span 4 (left) versus span 12 (right).

A particularly noteworthy observation emerges from parametric analysis of the animation in Figure 2: while increasing sps generally yields marginal improvements in MSE performance across the SNR range examined, this enhancement asymptotically approaches a limit and does not constitute convergence toward the theoretical ideal baseline (absent pulse shaping). This behavior underscores a fundamental trade-off inherent to practical pulse-shaping implementations: enhanced

spectral efficiency through reduced sps incurs amplified inter-symbol interference, whereas elevated oversampling ratios improve timing robustness at the cost of increased computational complexity and memory consumption.

The eye and constellation diagrams provide valuable insights into the behavior of the ideal OTA system under different SPS configurations. As the number of samples per symbol increases, the eye opening widens, indicating reduced inter-symbol interference and improved signal integrity. Similarly, the constellation points become more distinct and well-separated, reflecting better performance in terms of error probability and overall system reliability.

The span parameter controls the width of the pulse shaping filter, and a larger span results in a more gradual transition band, leading to a flatter MSE curve across different roll-off factors.

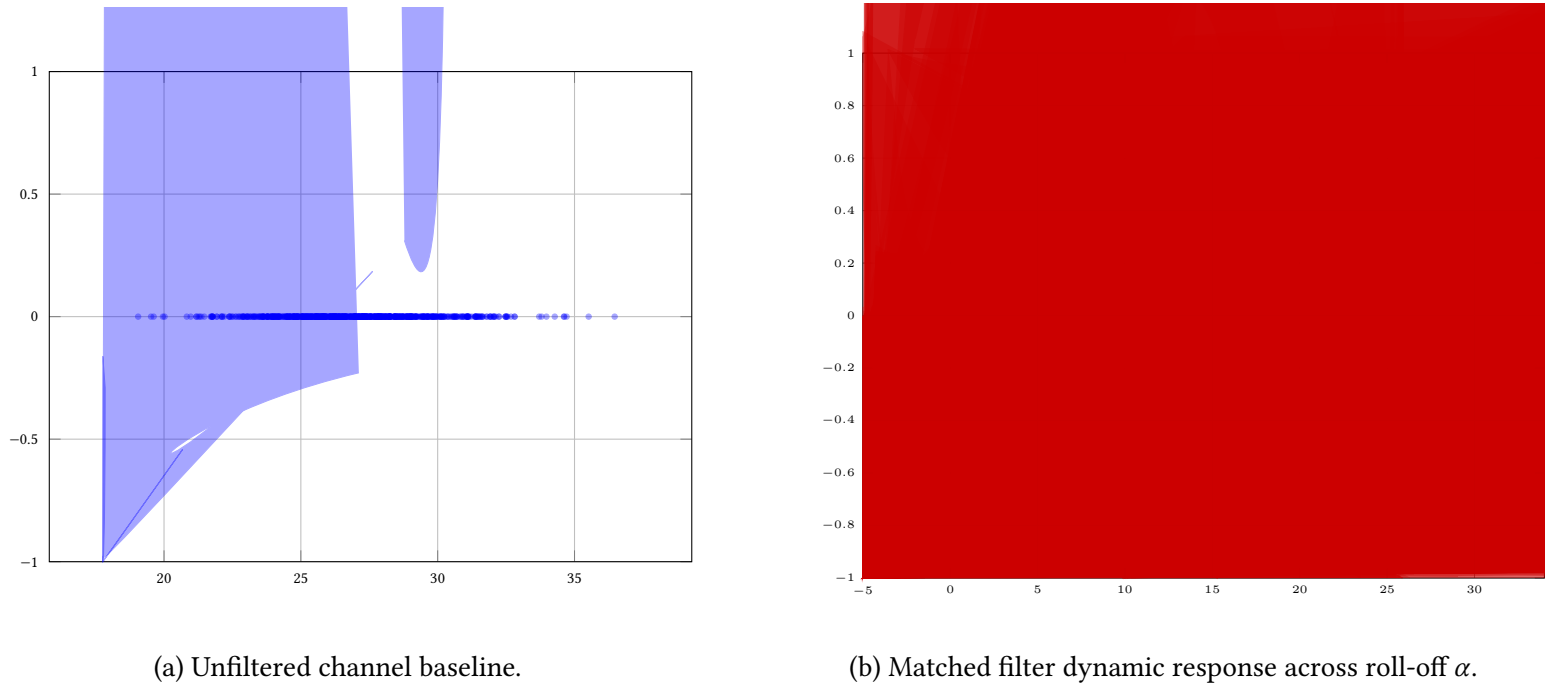


Figure 3: Constellation diagram comparison at SNR = 20 dB. Left: Unfiltered AWGN channel. Right: Animation demonstrating the suppression of truncation-induced ISI as α scales from 0.00 to 1.00 (span = 12, sps = 60).

The constellation diagrams in Figure 3 provide a visual representation of the received signal's integrity under both unfiltered and filtered conditions. The left subfigure illustrates the dispersion of constellation points in the absence of filtering, indicating potential symbol overlap and increased error probability. In contrast, the right subfigure demonstrates the effect of matched RRC filtering, where the constellation points are more tightly clustered, reflecting improved symbol separation and reduced inter-symbol interference. This visual evidence reinforces the quantitative findings from the MSE analysis, highlighting the critical role of pulse shaping in enhancing system performance.

What is particularly noteworthy is that the constellation diagram with RRC filtering exhibits a more compact and well-defined symbol distribution. This is because in the unfiltered scenario no upsampling takes place, and the symbols are transmitted at the symbol rate. Matlab measures the average power using Eq. (12). At 20dB, the noise variance is injected according to Eq. (13). The resulting standard deviation and average estimate are given by Eq. (6), Eq. (7), and Eq. (8).

$$\sigma_{\text{noise}} = \sqrt{2916} \approx 54 \quad (6)$$

$$\text{Estimated_Average} = \frac{\text{Ideal_Sum} + w}{K} = 27 + \frac{w}{20} \quad (7)$$

$$\sigma_{\text{Estimated_Average}} = \frac{\sigma_{\text{noise}}}{K} = \frac{54}{20} = 2.7 \quad (8)$$

In contrast, the RRC filtered scenario employs upsampling, so the RRC filter is normalized to unit energy Eq. (9). When the convolution happens with the upsampled signal, the energy is spread across the *sps* samples, resulting in a lower effective noise variance per sample. In greater detail, the noise variance after filtering is described by Eq. (10), and the standard deviation of the estimated average is given by Eq. (11).

$$\sum_n g[n]^2 = 1 \quad (9)$$

$$\sigma_{\text{noise, filtered}}^2 = \frac{\sigma_{\text{noise}}^2}{\text{sps}} = \frac{2916}{10} = 291.6 \quad (10)$$

$$\sigma_{\text{Estimated_Average, filtered}} = \frac{\sigma_{\text{noise, filtered}}}{K} = \frac{\sqrt{291.6}}{20} \approx 0.85 \quad (11)$$

$$P_{\text{signal}} = \frac{1}{N} \sum (\text{Ideal_Sum})^2 \approx 540^2 = 291600 \quad (12)$$

$$\sigma_{\text{noise}}^2 = \frac{P_{\text{signal}}}{100} = \frac{291600}{100} = 2916 \quad (13)$$

The eye diagram animation in Figure 4 illustrates the temporal evolution of the received signal's integrity as the number of samples per symbol varies. The left subfigure, corresponding to a truncated filter with span = 4, shows that as *sps* increases, the eye opening becomes wider, indicating improved timing margins and reduced inter-symbol interference. The right subfigure, with an extended filter of span = 12, demonstrates that even at lower *sps* values, the eye remains relatively open due to the longer filter length, which provides better pulse shaping and mitigates ISI effects. This visual evidence reinforces the quantitative findings from the MSE analysis, highlighting the critical role of both *sps* and filter span in enhancing system performance.

If we take a closer look near the top of each diagram 4 we see that the aggregated amplitude can take any real value, there are no voltage levels, and the eye diagram is not a square shape, but rather a continuous distribution of amplitudes. This is because the transmitted signals are analog values, and the superposition of these signals results in a continuous range of possible amplitudes at the receiver. The eye diagram reflects this continuous nature, showing the variations in amplitude over time and the impact of noise on the received signal. The eye opening indicates the time intervals where the signal is most stable and least affected by noise, which is crucial for accurate symbol detection in an analog OTA computation scenario.

Unlike digital slicers that require an open eye mask, OTA computation systems rely on the continuous nature of the received signal. In an ideal Nyquist system the derivative of every pulse passes through zero at the sampling instant, resulting in flat trajectory slopes at $t/T_s = 1.0$. In 4 the trajectories either reach a peak, a trough, or an inflection point where the slopes flatten out, in other words the derivative of the pulse is constant at the Nyquist strobe point 14.

$$\left. \frac{dp(t)}{dt} \right|_{t=0} = 0 \quad (14)$$

In OTA, timing margin is quantified by the curvature (second derivative) around the strobe instant 15. The bundle's tightness around $t/T_s = 1.0$ shows the receiver's sensitivity to clock drift: a wider, flatter plateau around $t/T_s = 1.0$ means small timing misalignments will not severely degrade the estimated mean.

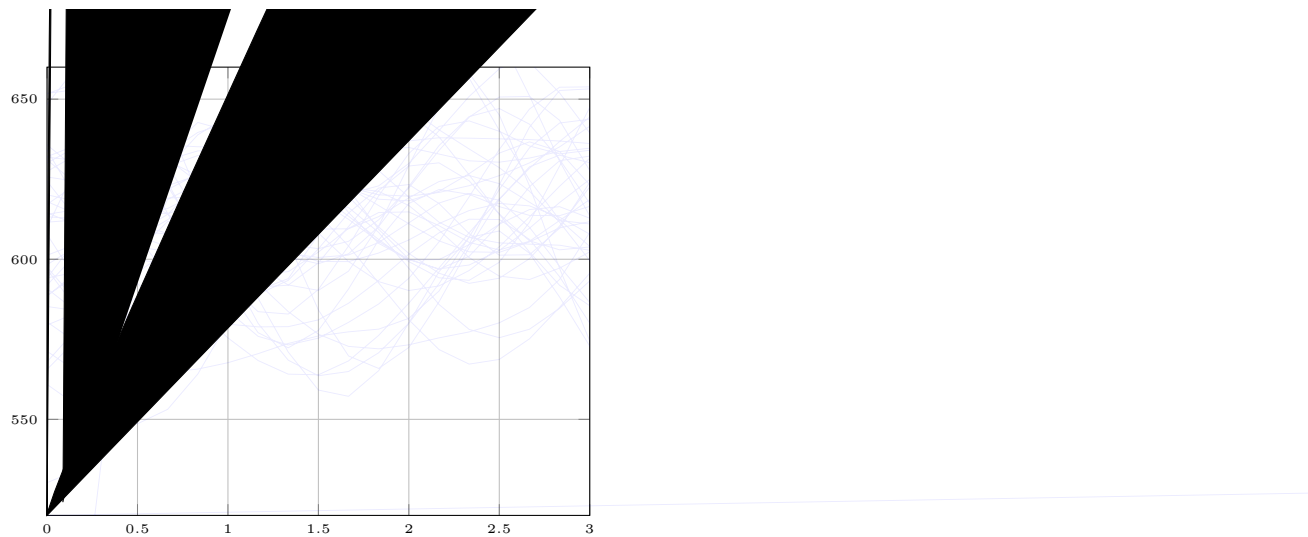


Figure 4: Eye diagram evolution for fixed roll-off $\alpha = 0.70$.

$$y(T_s + \Delta t) \approx y(T_s) + \frac{1}{2}y''(T_s)\Delta t^2 \quad (15)$$

1.c MATLAB Simulink Architecture

Transitioning from raw code to a block-level architecture, MATLAB Simulink provides a clear visual representation of the feedforward ideal system. In this model, the transmitters use standard PAM without the need for complex feedback loops or equalizers.

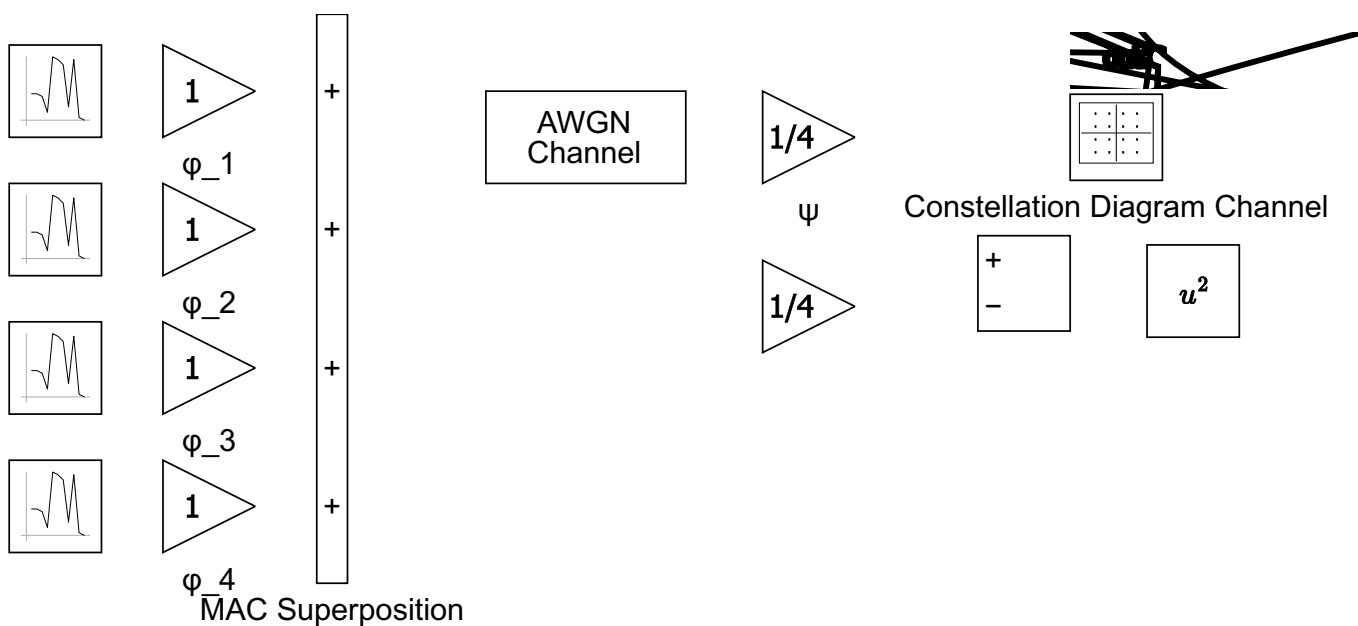


Figure 5: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

To bridge the gap between theory and practical system implementation, an enhanced model incorporating Raised Cosine pulse shaping is presented in Figure 6. This architecture employs matched filtering pairs at both transmission and reception stages, enabling quantitative assessment of performance degradation attributable to finite symbol-rate sampling and spectral roll-off constraints.

Inspection of the MSE calculation scope output demonstrates the quantitative deviation between the recovered aggregated estimate and the theoretical true value, reflecting the stochastic influence of the additive Gaussian noise. It is noteworthy that practical over-the-air computation mandates time-varying transmission (as opposed to static constant-valued signals),

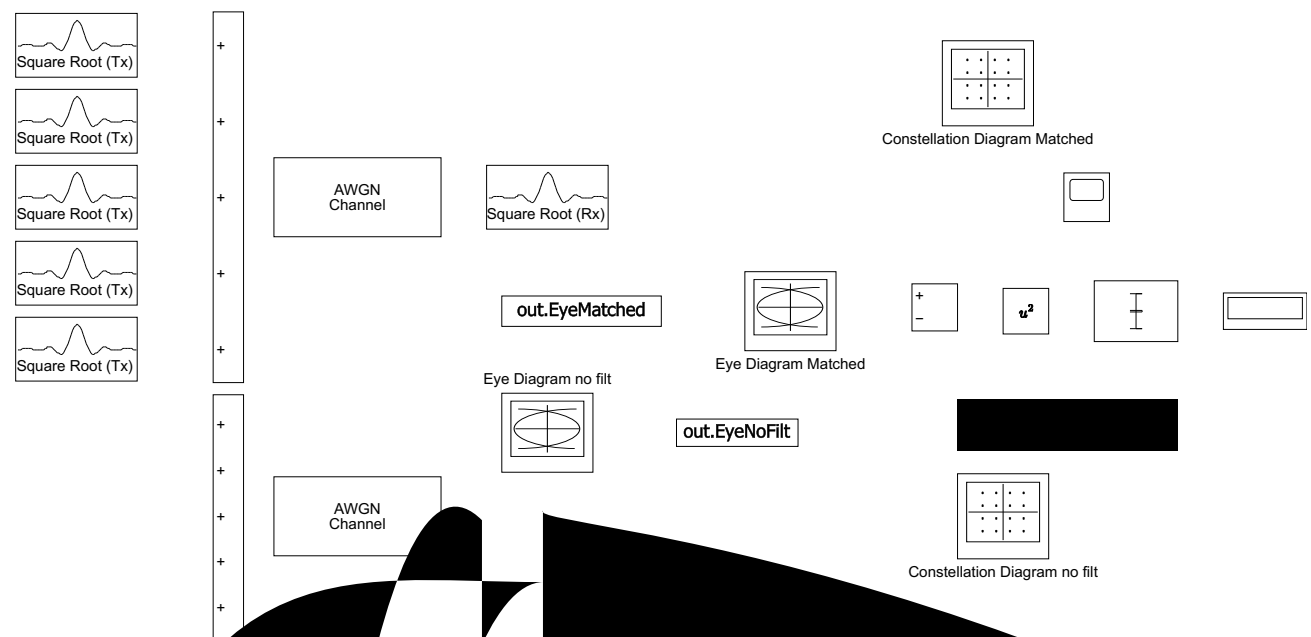


Figure 6: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

as pulse shaping through Finite Impulse Response (FIR) filtering inherently requires symbol-rate sampling and subsequent interpolation. Consequently, systems employing matched filter pairs necessitate carefully synchronized symbol generation to avoid spurious transient responses.

The constellation diagrams provide a clear visual confirmation of the role played by pulse shaping in the received signal. The plot corresponding to the fusion center observation after the matched RRC filter shows a more compact and well-defined symbol distribution, indicating that the filtering stage has improved temporal alignment and reduced the impact of inter-symbol interference. In contrast, the constellation associated with the theoretical receiver model without pulse shaping appears more dispersed and less structured, reflecting the absence of the smoothing and energy-concentrating effect provided by the matched filter. This comparison highlights that the matched RRC filter contributes not only to spectral shaping, but also to preserving symbol integrity and improving the reliability of the recovered OTA estimate.

When an RRC filter is generated using MATLAB/Simulink's underlying `rcosdesign` math, the filter is normalized for Energy, not Peak Amplitude. The algorithm guarantees that the sum of the squares of the filter taps equals 1.0 ($\sum h[n]^2 = 1$). However, in discrete-time Digital Signal Processing (DSP), when you push a signal through an interpolation filter, the energy gets spread out across the sps samples. When the signal passes through the Tx RRC filter, travels through the air, and passes through the Rx RRC matched filter, the continuous-time math is perfect, but the discrete-time convolution sum inherently scales the output by a factor related to the sps.

The eye diagrams further illustrate the impact of pulse shaping on the received signal. The eye diagram corresponding to the fusion center observation after the matched RRC filter shows a more open and well-defined eye pattern, indicating that the filtering stage has improved temporal alignment and reduced inter-symbol interference. In contrast, the eye diagram associated with the theoretical receiver model without pulse shaping appears more closed and less structured, reflecting the absence of the smoothing and energy-concentrating effect provided by the matched filter. This comparison highlights that the matched RRC filter contributes not only to spectral shaping, but also to preserving symbol integrity and improving the reliability of the recovered OTA estimate.

1.d GNU Radio Validation

Finally, the ideal model is verified using a SDR environment in GNU Radio. To maintain the ideal conditions mathematically defined in (2), the Root Raised Cosine (RRC) filters are perfectly matched. Specifically, the interpolation rate at the transmitters

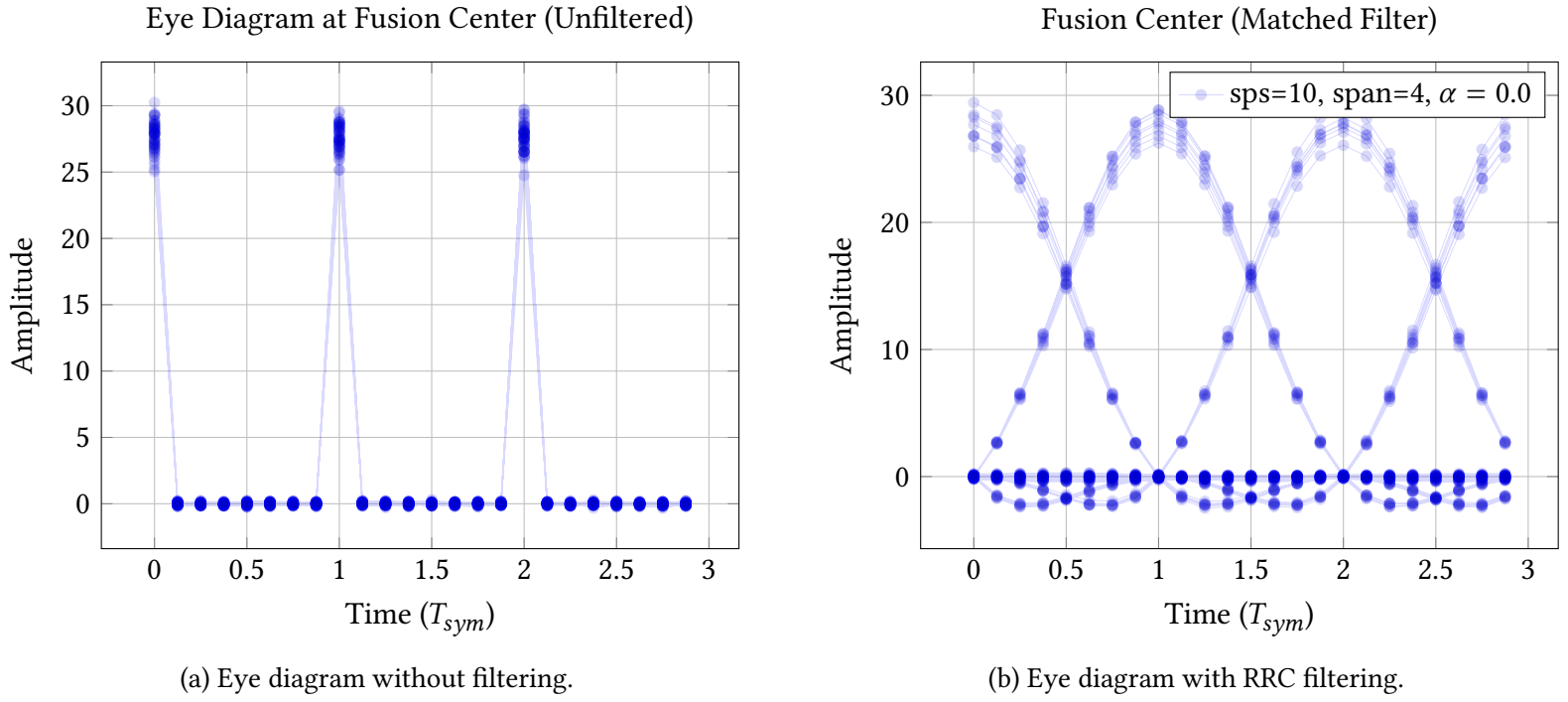


Figure 7: Eye Diagrams with and without filtering.

and the decimation rate at the receiver are identically matched to the symbol design rate (**sps_actual = sps_design = 6**). In other words the upsampling rate of the FIR filters matches the spacing between sample generation.

An important architectural consideration emerges from the GNU Radio implementation: in the absence of explicit digital modulation schemes, strategic downsampling at the transmitter becomes mandatory to enable proper filter operation. Specifically, each individual sensor transmits one symbol while simultaneously discarding sixteen interpolated samples, thereby maintaining compliance with Nyquist sampling criterion while leveraging the spectral efficiency benefits of pulse shaping.

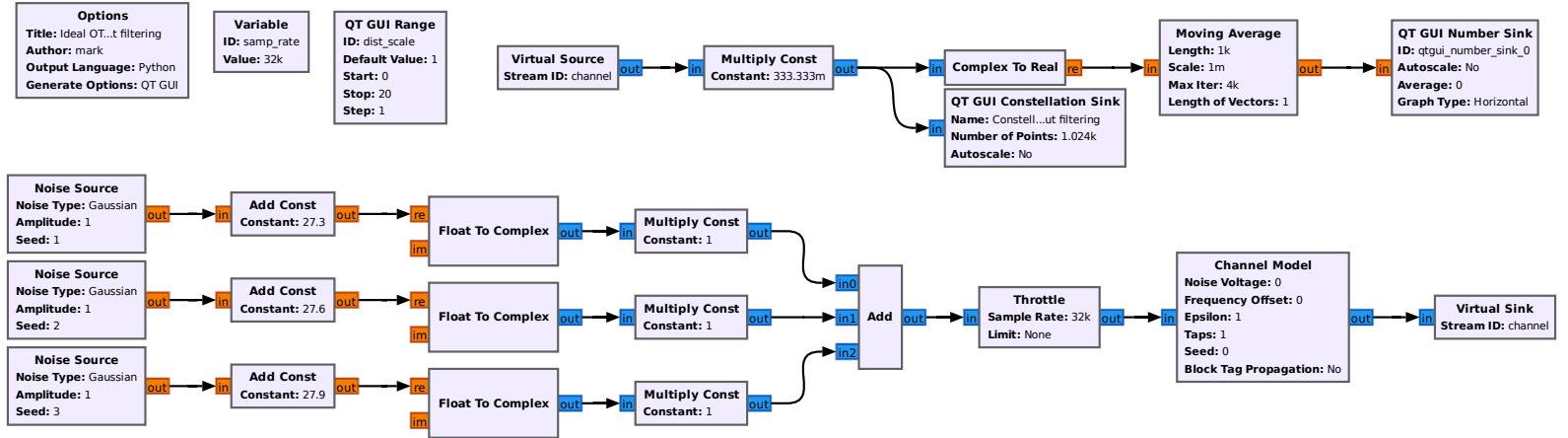


Figure 8: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

The matched filter perfectly recovered the symbol, but the amplitude fell short (26.4 instead of 27.4). The one-unit reduction in amplitude results from a fundamental property of Raised Cosine (RC) filters: when calculating MSE. When a RRC filter is generated using MATLAB/Simulink's underlying rcosdesign math, the filter is normalized for Energy, not Peak Amplitude. The algorithm guarantees that the sum of the squares of the filter taps equals 1.0 ($\sum h[n]^2 = 1$). However, in discrete-time DSP, when a signal passes through an interpolation filter, the energy is spread out across the SPS samples. When the signal passes through the Tx RRC filter, travels through the air, and passes through the Rx RRC matched filter, the continuous-time math is perfect, but the discrete-time convolution sum inherently scales the output by a factor related to the sps.

In commercial transceivers, the filters are not expected to perfectly preserve the raw voltage scale. Instead, a static scalar

multiplier (Gain block) is applied at the receiver, or an Automatic Gain Control (AGC) loop is used, to rescale the discrete digital values back to physical analogue units (such as Volts or Celsius) before the data reaches the Constellation Diagram or the Fusion Center average calculation.

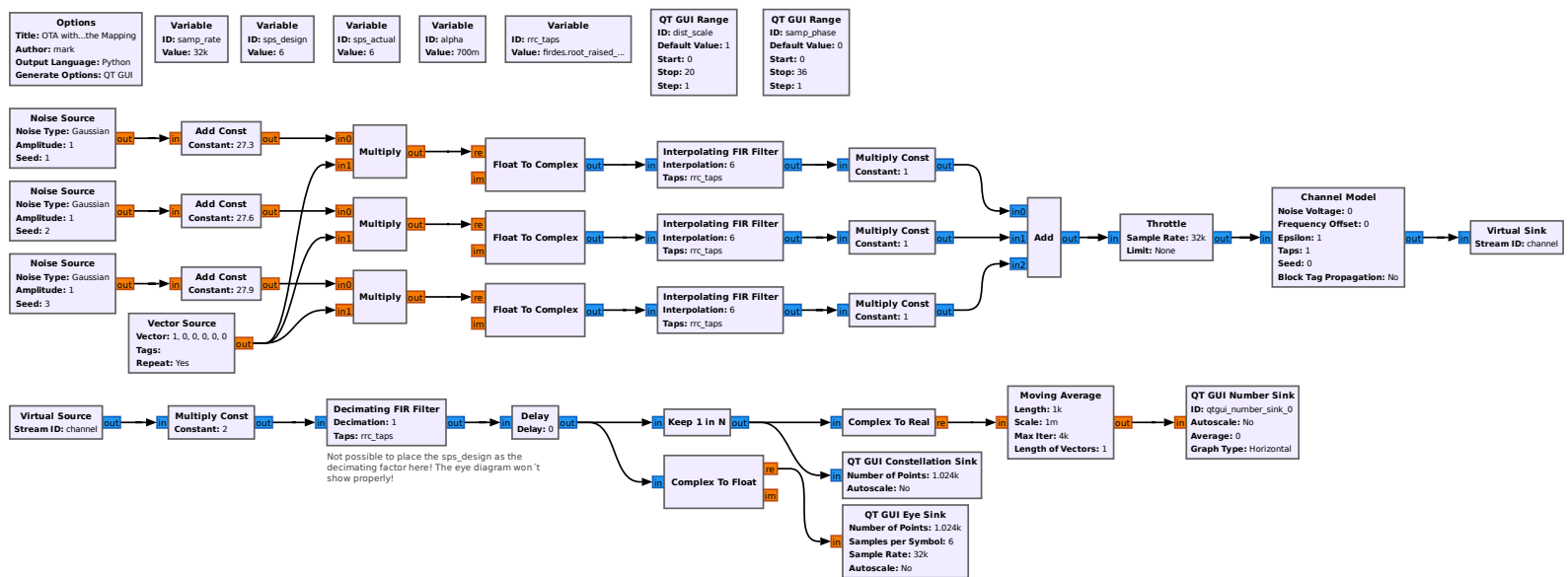


Figure 9: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

Under these constraints, the Eye Diagram remains perfectly "open" at the optimal sampling phase, and the Constellation Diagram exhibits tightly clustered points. The absence of horizontal dispersion in the constellation confirms that, prior to the introduction of timing offsets or Faster-Than-Nyquist signaling, the MAC superposition principle flawlessly computes the arithmetic average of the agricultural sensors.

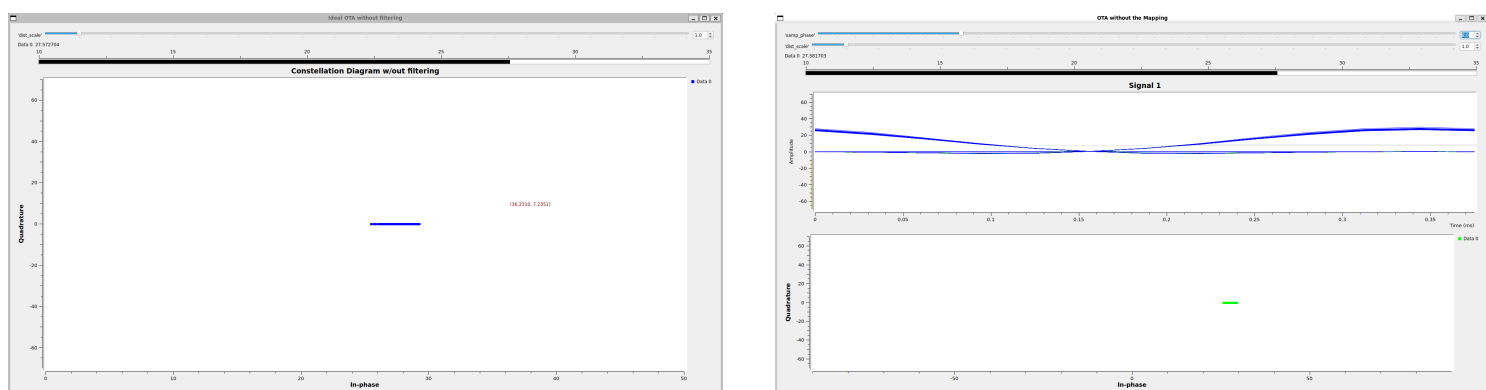


Figure 10: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

A critical implementation detail warranting emphasis concerns the optimal sampling phase selection. Empirical observation from the GNU Radio receiver indicates that maximum data recovery is achieved at a sampling phase offset of 33 samples out of the 96-sample cycle⁽¹⁾. This phase offset reflects the group delay inherent to the polyphase interpolating and decimating FIR filter implementations, necessitating careful calibration for optimal demodulation performance.

It is also useful to note that the GNU Radio implementation, while functionally equivalent to the MATLAB/Simulink model, introduces additional practical considerations, such as real-time processing constraints and hardware-induced latency. These factors may influence the observed performance metrics, particularly in scenarios where the system is subjected to dynamic channel conditions or when operating near the limits of the hardware's processing capabilities. Therefore, while the idealized models provide a valuable theoretical benchmark, real-world deployments must account for these additional complexities to ensure robust and reliable operation.

¹The fundamental cycle spans 16 symbols; with 6 samples per symbol, the total cycle length is $16 \times 6 = 96$ samples.

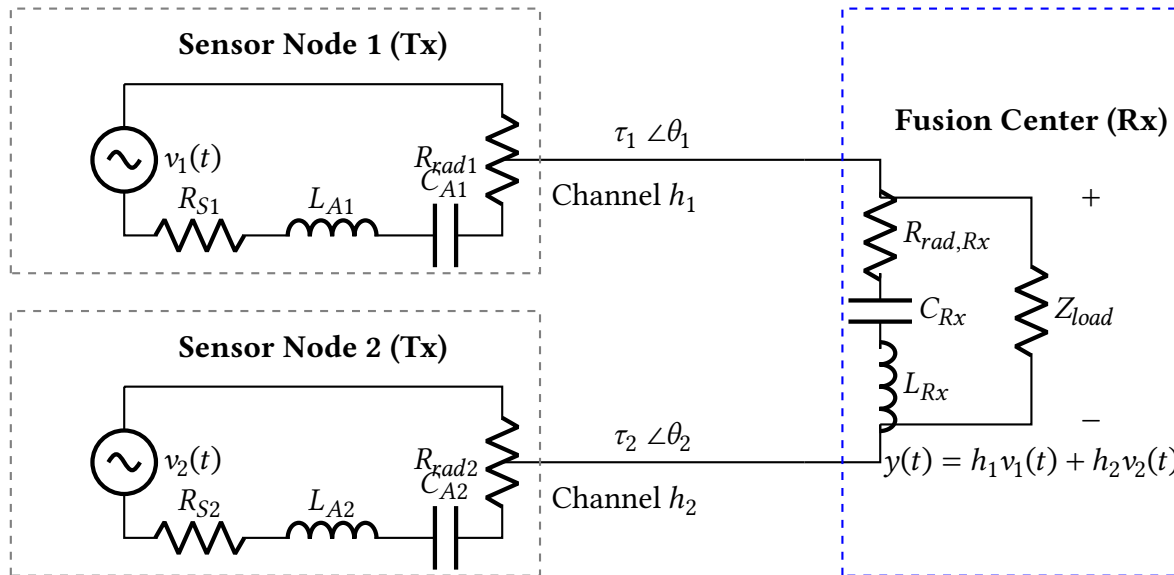


Figure 11: Circuit equivalent of the ideal OTA computation network. Each sensor node is modeled as a voltage source with internal impedance, and the fusion center aggregates the received signals through a common load. The wireless channels are represented as transmission lines with associated delays and phase shifts, capturing the effects of propagation in a simplified manner.

The circuit equivalent of the ideal OTA computation network is depicted in Figure 11. This representation abstracts the wireless channel into a simplified electrical model, where each sensor's transmission is treated as a voltage source, and the fusion center aggregates these signals through a common load. The circuit

2 System Model and Practical Challenges in OTA

A system that utilizes OTA computing can be modeled in various ways depending on the specific use case and propagation environment. Most architectures involve a set of EDs that transmit concurrently to a FC or Edge Server (ES). The communication channel is a critical component of this model, as it introduces physical impairments such as fading, noise, and path loss. In addition to the channel model, practical challenges inherent to wireless communications—such as time sampling errors, multipath propagation, and waveform design—can significantly degrade the computational reliability of OTA systems. To mitigate these effects, optimization strategies including power management, transceiver design, and equalization must be employed.

2.a System Architecture and Channel Model

To mathematically formulate the impact of channel impairments on OTA computing, we consider a discrete-time network model consisting of multiple EDs deployed in an agricultural environment. These devices utilize PAM to transmit local temperature readings to a central FC.

Let $x_i[n]$ denote the discrete-time signal transmitted by the i -th sensor, as defined in Eq. (16). For an illustrative scenario involving two sensors, the transmitted sequences are defined as:

$$x_i[n] \text{ denotes the signal transmitted by sensor } i \quad (16)$$

$$x_1[n] = [2, -1, 1] \quad (17)$$

$$x_2[n] = [1, 2, -1] \quad (18)$$

2.a.i The Ideal Target Function

The primary objective of OTA computing is to leverage the multiple access channel (MAC) superposition property to aggregate data natively in the air. Assuming a flat-fading, noise-free, and perfectly synchronized channel, the ideal target function received at the FC, denoted as $s[n]$, is the direct sum of the transmitted signals:

$$s[n] = \sum_{i=1}^K x_i[n] \quad (19)$$

where K is the total number of transmitting devices. For the given sensor data, the ideal aggregated sequence is evaluated as Eq. (20).

$$s[n] = [3, 1, 0] \quad (20)$$

2.a.ii Multipath Channel Distortion

In practical agricultural deployments, the physical environment exhibits multipath characteristics (e.g., reflections from terrain or machinery). This is modeled as a Finite Impulse Response (FIR) filter. Assuming the three-tap channel response in Eq. (21), the actual received signal $y[n]$ is the convolution of the ideal aggregated signal with the channel response:

$$h[n] = [1.0, 0.5, -0.2] \quad (21)$$

$$y[n] = s[n] * h[n] = \sum_{k=0}^n s[k]h[n-k] \quad (22)$$

Evaluating this convolution yields the distorted sequence in Eq. (23). The extension of the sequence length from three to five symbols mathematically illustrates the presence of ISI, wherein previous and future symbols bleed into the current sampling instance.

$$y[n] = [3.0, 2.5, -0.1, -0.2, 0.0] \quad (23)$$

2.a.iii Mean Squared Error (MSE) Evaluation

To quantify the degradation caused by the multipath channel, the MSE is calculated between the ideal target function $s[n]$ and the actual received signal $y[n]$. The error at each discrete time step is given by Eq. (24). For a sequence of length N , the overall MSE is defined as:

$$e[n] = y[n] - s[n] \quad (24)$$

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} (y[n] - s[n])^2 \quad (25)$$

For the evaluated sequences, the discrepancy between the ideal sum and the ISI-corrupted signal results in an $\text{MSE} = 0.46$. This theoretical error establishes the baseline challenge that must be addressed through waveform design or digital equalization techniques at the FC.

2.a.iv Matlab Simulation of Multipath Effects

The following Matlab code snippet simulates the multipath channel effects on the transmitted signals and computes the resulting MSE. The code defines the transmitted sequences, the channel impulse response, and performs the convolution to obtain the received signal. Finally, it calculates the MSE between the ideal and received signals.

Listing 3: Matlab Simulation of Multipath Channel Effects and MSE Calculation

```

1
2      ...
3
4      % Fractional Delay System Object
5      fracDelay = dsp.VariableFractionalDelay('MaximumDelay', 100);
6
7      % Raised Cosine Filter
8      rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");
9
10     % Aggregate noiseless transmitted signal
11     % Initialized as a column vector to prevent MATLAB matrix expansion bugs
12     rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);
13
14     for k = 1:K
15         % The Air acts as an ideal adder (MAC Superposition)
16         tx_sig = conv(txUpSamped(:,k), rcFilter(:)) ;
17
18         % Channel: Multipath fractional delay
19         %  $h(t) = d(t) + 0.5*d(t - 0.25T_s)$ 
20         % channel_out = conv(tx_sig, h_channel, "same");
21
22         delayed_sig = fracDelay(tx_sig, delay_frac * sps_val);
23
24         % Force rcFilter to column vector '(:)' for safe 1D convolution
25         rx_agg_signal = rx_agg_signal + tx_sig + 0.5 * delayed_sig ;% eyediagram
26         (node_1_tx, sps_val)
27     end
28     ...

```

In the above code, the 'dsp.VariableFractionalDelay' object is used to introduce a fractional delay to simulate multipath effects. The 'rcosdesign' function generates a square-root raised cosine filter to shape the transmitted signals. The received signal is computed by convolving the transmitted signals with the channel response, and the MSE is calculated to quantify the distortion introduced by the multipath channel.

The MSE value differs significantly from the ideal case, highlighting the importance of addressing multipath effects in practical OTA computing systems.

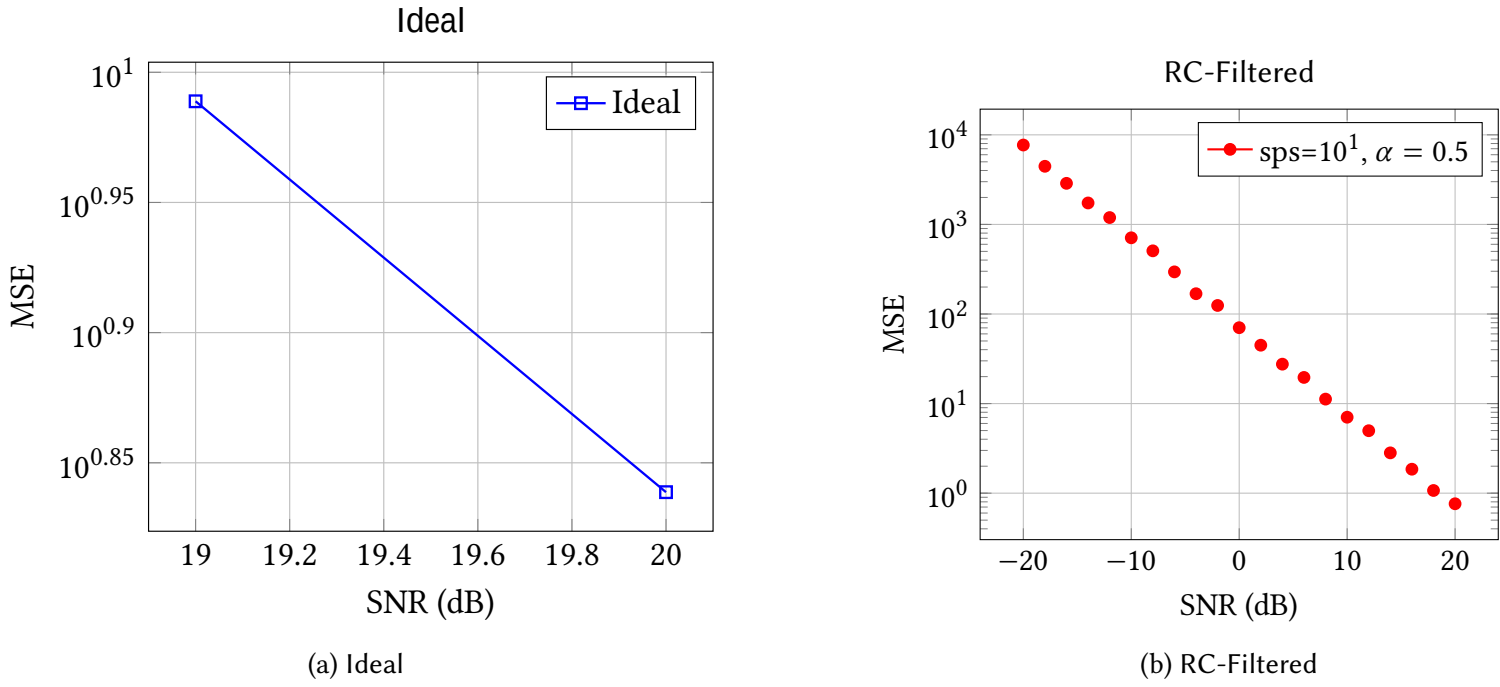


Figure 12: Comparison of MSE performance between the ideal baseline and the Raised Cosine filtered scenario.

2.b Practical Challenges

2.b.i Channel State Information (CSI) Imperfection

The theoretical model assumes perfect knowledge of the channel impulse response. In practice, the need for accurate and fresh CSI at transmitters or receivers is critical for precoding and equalization techniques [2], [6], [8]. Obsolete or noisy CSI severely limits the ability to pre-distort signals, exacerbating the MSE.

2.b.ii Waveform and Filter Design

As demonstrated by the sequence expansion in the multipath model, standard digital communication transceivers suffer from time sampling errors and ISI. Designing custom pulse shapes, such as optimized Root Raised Cosine (RRC) filters or DNN-generated waveforms, is essential to force zero-crossings at the exact sampling instances, thereby suppressing ISI natively over the air [2].

2.c Optimization Strategies to Enhance OAC Performance

To mitigate the aforementioned challenges, several optimization frameworks have been proposed in the literature:

- **Power Management:** Strategies to mitigate aggregation errors involve optimizing power control policies at the EDs to align received signal amplitudes, thereby minimizing the overall MSE or maximizing the convergence speed of federated learning algorithms [1], [2], [6].
- **Transceiver Design:** The joint design of pre-processing functions at the transmitter and post-processing equalizers at the receiver allows the system to match the channel characteristics to the desired nomographic function, effectively canceling multipath echoes and minimizing computational distortion [4], [5], [6], [8].

3 Experimental Methodology and Implementation

3.a Analog OAC and Nomographic Functions

To validate the mathematical models established in the previous section, an experimental framework was developed. The implementation focuses on analog OTA to compute nomographic functions—specifically, the arithmetic mean of distributed temperature sensors. By leveraging the superposition property of the multiple access channel, the network natively aggregates uncoded sensor readings in the air, transforming the physical medium into a computational processor.

3.b GNU Radio System Architecture

To physically simulate the analog OTA network, a Software Defined Radio (SDR) architecture was constructed using GNU Radio. The simulation environment accurately models the concurrent transmission of multiple Edge Devices (EDs) to a central Fusion Center (FC).

3.b.i Hardware and Software Parameters

The baseline configuration for the SDR implementation is detailed in Table 2. These parameters were selected to balance computational load while providing sufficient bandwidth to observe channel impairments.

Table 2: Key SDR Hardware and Software Parameters for the OAC Implementation

Parameter	Value	Description
<i>Hardware Configurations</i>		
Center Frequency (f_c)	2.4 GHz	ISM band, selected to minimize external interference.
Sample Rate (f_s)	32 kHz	Defined baseband rate for the SDR sink/source blocks.
Antenna Configuration	SISO	Single-Input Single-Output for initial baseline testing.
<i>Software & Baseband Processing</i>		
Modulation Scheme	PAM	Pulse Amplitude Modulation for analog temperature transmission.
Samples per Symbol (sps)	6	Determines the interpolation/decimation rate for the signal.
Pulse Shaping Filter	RRC	Root Raised Cosine filter applied to mitigate ISI.
Roll-off Factor (α)	0.3	Standard excess bandwidth factor defining the filter's frequency transition band.

3.b.ii Pulse Shaping and Channel Impairments

Two critical components of the GNU Radio flowgraph govern the physical behavior of the transmission:

- **Root Raised Cosine (RRC) Filter:** To constrain the transmission bandwidth and mitigate Intersymbol Interference (ISI), RRC filters are employed for pulse shaping at both the transmitters and the receiver. The RRC filter ensures that, under ideal Nyquist sampling rates, the overlapping tails of adjacent symbols evaluate to zero, thereby preserving the integrity of the target function computation.

- **Channel Model:** To accurately emulate an agricultural deployment, a Channel Model block is introduced to inject realistic wireless impairments. This block models Additive White Gaussian Noise (AWGN) and user-defined multipath delay taps, which scatter the signal arrivals and intentionally induce ISI to test the system's robustness.

3.b.iii Flowgraph Configurations

The network was simulated under two primary conditions. Figure 13 illustrates the continuous-time analog simulation where sensor data is continuously evaluated. Figure 14 depicts the "Faster-Than-Nyquist" (FTN) scenario, wherein the transmission rate is artificially accelerated to break the Nyquist limit and observe the resulting ISI degradation.

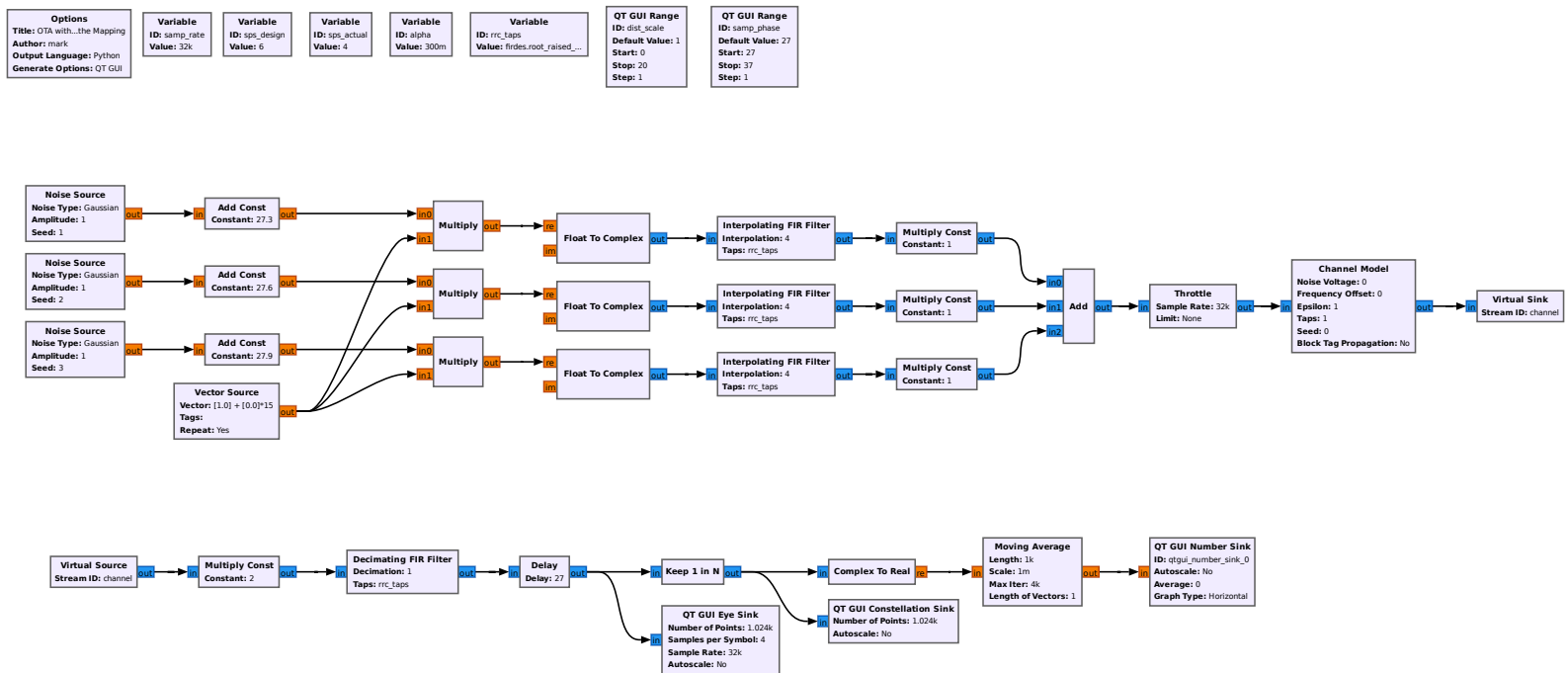


Figure 13: GNU Radio Companion flowgraph for the OTA implementation including Channel Model and RRC filters.

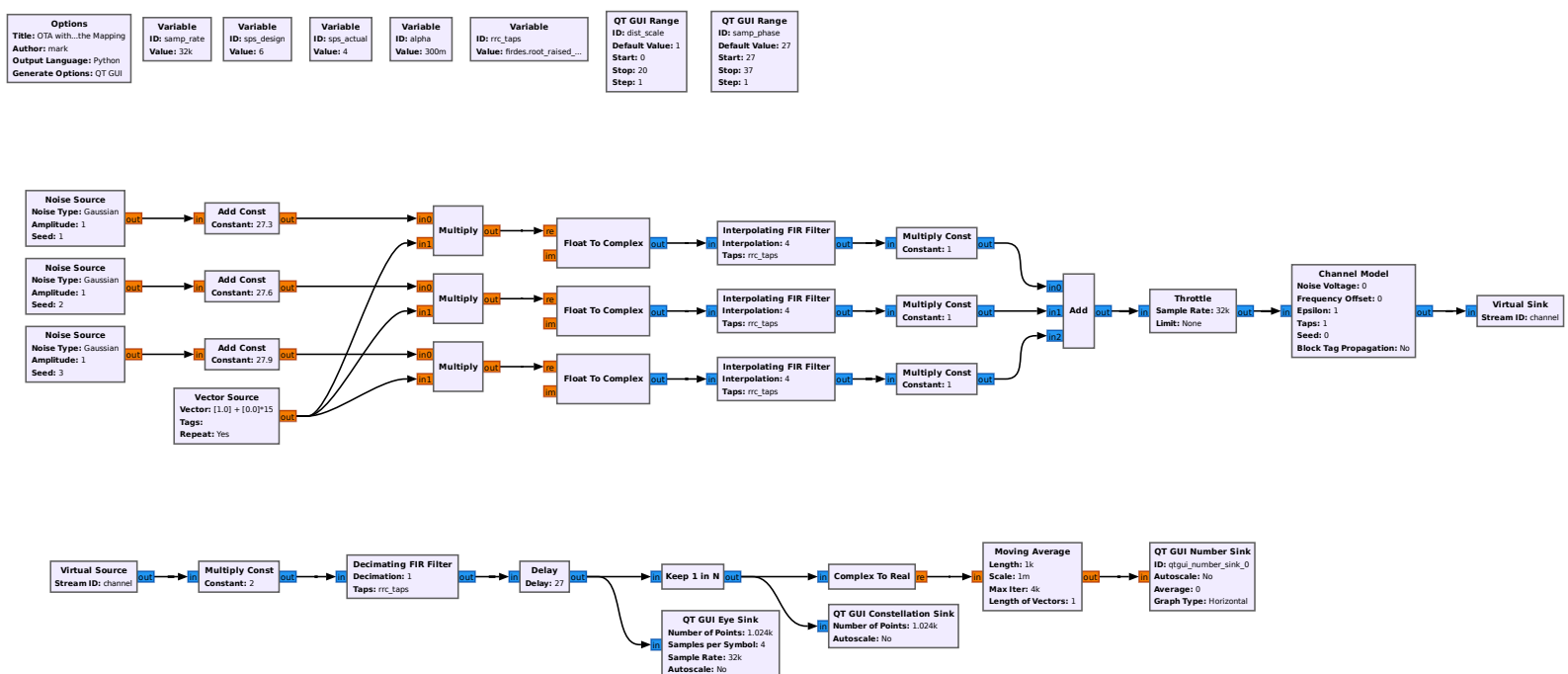


Figure 14: GNU Radio Companion flowgraph configured to induce Intersymbol Interference (ISI) by mismatching the design and actual sampling rates.

3.c MATLAB and Simulink Integration

While GNU Radio provides a robust framework for real-time SDR execution and analog signal evaluation, future phases of this methodology will incorporate MATLAB Simulink. Simulink will be utilized to model advanced feedback loops, custom waveform generation, and digital equalizers (e.g., Minimum Mean Square Error (MMSE) receivers) that are mathematically complex to implement in standard GNU Radio environments. This dual-software approach allows for both practical SDR validation and rigorous mathematical algorithmic design.

4 Experimental Results and Discussion

4.a Validation of the Implemented OAC Network

Demonstration of successful signal superposition and computation via the SDR platform[3], [6]. Comparison of achieved computation result against the ideal theoretical value[6], [9].

4.b Performance Evaluation under Impairments

Quantifying the performance metrics, such as Mean Squared Error (MSE) or Computation Error Rate (CER), of the implemented scheme[2], [6]. Analysis of the impact of deliberately introduced synchronization errors or varying channel noise (SNR) on the computation accuracy[2], [6]. Comparison with Separation Schemes (Simulated Baseline); Illustrating the resource savings and computation rate achieved by the OAC approach relative to traditional orthogonal access (e.g., Time-Division Multiple Access (TDMA))[3], [4], [6].

5 Conclusion and Future Work

5.a Summary of Thesis Findings

Recap of the feasibility and performance demonstrated by the SDR-based OAC network.

5.b Future Research Directions

Exploring more complex OAC architectures, such as multi-antenna systems Multiple-Input Multiple-Output (MIMO) or using Reconfigurable Intelligent Surfaces (RIS)[2], [10]. Further investigation into standardized OAC protocols for compatibility with modern wireless networks (e.g., LTE/5G protocols)[6], [9].

A MATLAB Code for OTA Simulation

This appendix contains the core MATLAB scripts used to generate the simulation data discussed in Chapter 2. Note that we utilized the Raised Cosine method [6].

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